

PAPER 1 / QUESTION 95

28 October 2020 / TZ0 / HL Paper 1 / original question 6 / 8820-6501

95. An object of mass $2m$ moving at velocity $3v$ collides with a stationary object of mass $4m$. The objects stick together after the collision. What is the final speed and the change in total kinetic energy immediately after the collision?

	Final speed	Change in total kinetic energy
A.	v	$3 mv^2$
B.	v	$6 mv^2$
C.	$2v$	$3 mv^2$
D.	$2v$	$6 mv^2$